Joanna Hoang

Forward Deployed & Solutions Engineering | Applied AI, Integrations, & Deployment
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PROFESSIONAL SUMMARY

Customer-facing engineer with 5+ years building, integrating, and deploying technical systems in customer environments: federal AI identity and security programs at NEC, enterprise AI enablement at Intel, and freelance AI application builds. Takes work from discovery and scoping through hands-on build, field testing, deployment, and support. Previously adjudicated Top Secret clearance (inactive).

SKILLS

Engineering: Python, SQL, PostgreSQL, TypeScript, React, TanStack Start, Supabase, Bash, PowerShell, Git, Claude Code, Codex
Integrations & Identity: REST APIs, OpenAPI, JSON, webhooks, OAuth2, SAML SSO, JWT, mutual TLS, Postman, Tavern
Deployment & Security: AWS (EC2, S3), Azure, Cloudflare Workers, Docker, CI/CD, Linux, Trivy, NIST, CMMC 2.0, FedRAMP
AI & Data: LLMs, RAG, NLP, computer vision, biometric matching, PyTorch, TensorFlow, SageMaker, Grafana/PromQL
Customer-Facing: discovery, scoping, POCs and pilots, field testing, demos, onboarding, training, technical writing

EXPERIENCE

Kiernan Group Holdings Mar 2026 – Present
Forward Deployed Engineer (Freelance) Remote

- Built a six-section AI investment-research copilot demo (investment-intelligence.si) on public and simulated data, running portfolio math in deterministic code so the model could not invent figures; the client owned scoping and data.
- Delivered the POC in ~2 weeks as a solo engineer with Claude Code and Codex on React, TanStack Start, Supabase, and Cloudflare Workers, adding a streaming API endpoint, Zod validation, usage limits, and tests.
- Rebuilt the website for this risk and resilience firm serving government and law enforcement (kghpreview.com), restructuring a decade of content into 9 routes and scoping a front-end-only first version for an early direction decision.

NEC National Security Systems Apr 2023 – Feb 2026
Software Solutions Engineer (Federal Contractor) Arlington, VA (Remote)

- At this federal biometrics and physical-security integrator, owned the technical side of ~5 concurrent engagements in AI identity, access, and perimeter security for DHS, Secret Service, and USMS, from discovery and integration through field testing, deployment, and support on mid-six to multi-million dollar programs.
- Ran ~9 months of field testing with the customer on a federal biometric-capture pilot; when matching failed in the field, traced it to environmental lighting through same-day testing, proving the root cause and advancing the deal to pilot rollout.
- Led discovery that redirected a secure government facility from facial recognition to fiber-optic perimeter intrusion detection; built the real-time operator dashboard (alerts, activity type, timelines, live video), coordinated installation, and delivered post-sales support.
- Deployed systems in secure federal environments for USMS and the Secret Service, authoring Bash and PowerShell install and database scripts, verifying records end to end with SQL, and writing administrator and end-user guides.
- Designed and validated integrations between biometric engines, cameras, watchlists, & access-control systems (REST, webhooks, OAuth2); authored an OpenAPI 3.0 contract with vetting provider covering 12 scenarios, async callbacks, JWT, and mutual TLS.
- Wrote database migration scripts preserving transaction history through an enterprise software upgrade planned for 124 field offices, validating integrity through diff comparisons and required-field audits.
- Led 50+ discovery and requirements sessions, 25+ tailored demos, and 20+ stakeholder briefings and pilot readouts; owned technical RFP responses and delivered 10+ onboarding and training sessions.

Intel Corporation Jun 2022 – Mar 2023
AI Solutions Engineer Hillsboro, OR

- Delivered 30+ technical demos, workshops, and POC-style sessions helping enterprise developers in automotive, financial services, healthcare, and other verticals adopt AI workloads, including discrete-GPU demos at Intel Innovation 2022.
- Built and managed AWS EC2 environments for PyTorch and TensorFlow labs at Intel Innovation and developer webinars (hundreds of participants), including a two-node distributed-training lab and infrastructure for a 100+ developer AI hackathon.
- Validated a ResNet-50 transfer-learning workflow, Docker image, and conda environment before publication in Intel's AI workflows catalog; verified MLPerf benchmark results on AWS EC2.

Seeq Corporation Jun 2021 – Dec 2021
Full Stack Software Engineering Intern (MECOP Co-op) Seattle, WA (Remote)

- At this industrial analytics SaaS company, built a Date Shared feature end to end — added a timestamp column to the access-control table, propagated it through the REST API, and displayed it in the React UI.
- Converted AngularJS components to React, resolved API bugs, and wrote unit and integration tests (JUnit, Jest, Cucumber).

Portland General Electric Jun 2020 – Dec 2020
Data Science Intern (MECOP Co-op) Portland, OR

- Designed the back end (Node.js, Socket.IO, AWS EC2) for a multiplayer internal training tool on company values, owning stack selection and budget proposal; trained the support team at handoff.
- Reduced test loss 40%+ vs. a convolutional autoencoder using PyTorch transformer models for energy classification (e.g., EV detection for grid planning); built a reusable data-annotation tool (with usage docs) and handled data preprocessing.

EDUCATION

University of Portland B.S. in Computer Science